Tobias Mwangi-Hartley

Data Scientist | Applied Machine Learning

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PROFILE

Data scientist with 6 years of experience in analytics and data science, including more than 4 years in data science roles. Builds and evaluates forecasting and customer-propensity models, develops feature-engineered workflows in Python and SQL, and translates business needs into practical analytical tools. Experience includes scheduled model scoring and forecasting workflows using BigQuery and Vertex AI Workbench, a Dockerized Streamlit experiment tool, and automated tests for forecasting feature code. Seeking to develop further in machine learning engineering and production systems.

TECHNICAL SKILLS

Python (pandas, scikit-learn, LightGBM, XGBoost, statsmodels, pytest), SQL (BigQuery), dbt, Git, Docker, Streamlit, R (tidyverse, brms), Vertex AI Workbench, Looker, Jupyter

Machine learning: Feature engineering, forecasting, churn propensity modelling, model evaluation, SHAP

Statistics: A/B testing, sequential testing, causal inference, Bayesian statistics, time-series forecasting

EXPERIENCE

Data Scientist | Thistlebox (meal-kit subscriptions), Edinburgh

Feb 2022–Present

- Built a weekly, six-week-ahead demand forecasting workflow by recipe and region using LightGBM and calendar, price and menu-position features. The scheduled Vertex AI Workbench notebook writes forecasts to BigQuery for operations reporting in Looker; WAPE fell from 23% to 14%, with estimated food waste reduction of 9%.
- Developed an XGBoost churn propensity model scored weekly in BigQuery, with retention offers targeted to top-decile customers. A 10% holdout validated a 2.1 percentage-point reduction in churn in the treated group.
- Design and analyse approximately 40 A/B tests each year for growth and product teams; built a sequential testing calculator used by around 25 colleagues, writing the Python Streamlit app and its Dockerfile. The data platform team manages its Cloud Run deployment.
- Write production SQL in BigQuery and maintain dbt models for the analytics layer.
- Since 2025, have written pytest unit tests for forecasting feature code, reaching approximately 60% coverage.
- Present results to leadership monthly and delivered the internal “Stats for PMs” training three times.

Junior Data Analyst | Firth Energy Retail, Glasgow

Sep 2020–Jan 2022

- Performed customer segmentation and tariff analysis using R and SQL.
- Automated monthly regulator reports with R Markdown, saving the team about two days each month.

SELECTED PROJECT

Image Classification Course Project | Personal, 2025

- Fine-tuned a small CNN in PyTorch on a public food-images dataset as part of an online deep learning course; achieved approximately 80% accuracy. Learning project.
- Tools: PyTorch, torchvision, Google Colab.

EDUCATION

MSc Statistics, University of the Lothians | 2020 | Distinction

Dissertation: Bayesian hierarchical models for small-area energy demand

BSc Mathematics, University of the Lothians | 2019

LANGUAGES

English (native) | Swahili (conversational)